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Review Article

A comprehensive review of the influence of physicochemical properties of biodiesel on combustion characteristics, engine performance and emissions



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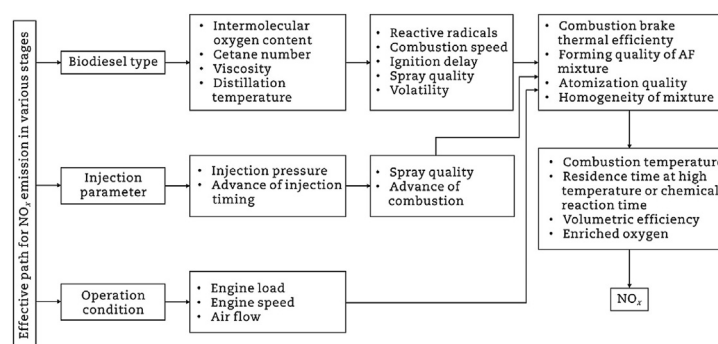
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HIGHLIGHTS

- Biodiesel is the most promising alternative fuel in CI engines.
- The primary source of the third-generation biofuel is the algae.
- Ignition delayed due to the occurrence of the physical and chemical processes.
- Summarized physicochemical properties of different biodiesels from first to third generation.

GRAPHICAL ABSTRACT



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ABSTRACT

Energy is a basic requirement for development in the world. The continuously rising demand for energy resources and the fast depletion of fossil fuel is raises researcher's

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concern to focus on alternative sources of energy that can replace the shortage of fossil fuels soon. Biodiesel is recognized as one of the potential alternative renewable energy fuels that can be easily available in a wide range in every part of the world. This paper reviews biodiesel's prospect and focuses on the different fuel properties (physicochemical) of 100 biodiesels from first-, second- and third-generation followed by the evaluation of CI engine characteristics. It has been observed that the fuel properties of first-, second- and third-generation biodiesel are compatible with the ASTM standards. The present paper discussed the potential of various generations of biodiesel feedstocks from production perspectives. This paper also examined many aspects of these feedstocks which include different biodiesel feedstock sources, biodiesel conversion technology, and second-generation biodiesel performance and emission characteristics. Most of the studies showed that biodiesel derived from plant feedstock is costly than diesel. Thus, more influence is to be given to non-edible sources. However, the engine characteristics are shown promising behavior with biodiesel and a slight increase in NO_x is also reported on using various biodiesel.

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1. Introduction

1.1. Present energy scenario

The rapid increase in energy consumption is due to the regular rise in automobiles' industrial and population growth and economic development. Fig. 1 shows the current energy scenario and the projection of energy consumption until 2035. Oil fuel is the dominant provider empowering the global economy. It is almost sure that diesel will acquire approximately 60% of the energy growth accounting for almost 80% of total energy supply in 2035 (Chhabra et al., 2021). The energy scenario is predicted to continue adding in the upcoming years (Yusri et al., 2018). Fig. 2 shows the prediction of per capita renewable water birthplaces availability ($\text{m}^3/\text{person}/\text{year}$) in the world. Further, future

utilization of activity in ceases of vigorous agricultural going-on to no small extent in the world by 70% (renewable water harvesting) and 30% only for high-income countries (Hajjari et al., 2017).

The above stats show that the world will be facing enormous problems for fulfilling the demand for energy resources for society, industries and other sectors in upcoming years. This increasing demand for energy has forced various researchers in the world to work for alternative sources of energy to meet the demand (Kumar Shukla et al., 2020; Palash et al., 2015). The burning of fossil fuels also produces harmful emissions polluting the environment. The transport sector is recognized as the most energy-consuming sector. To meet the increased demand for energy and reduce CO_2 emission, the world needs to have more effective and sustainable energy sources. Thus, the use of new clean-

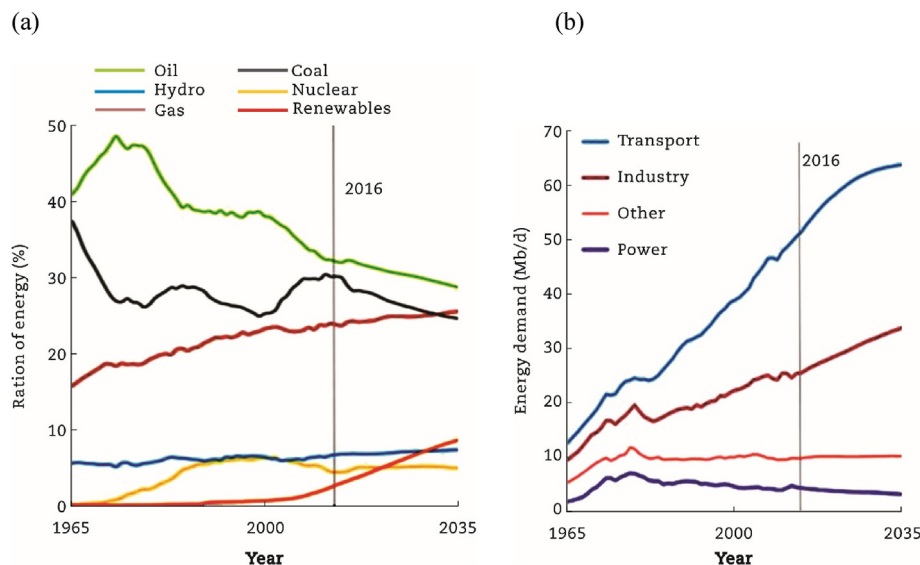


Fig. 1 – Scenario of energy consumption. (a) Shares of primary energy. (b) Liquids of demand by sector (Yusri et al., 2018).

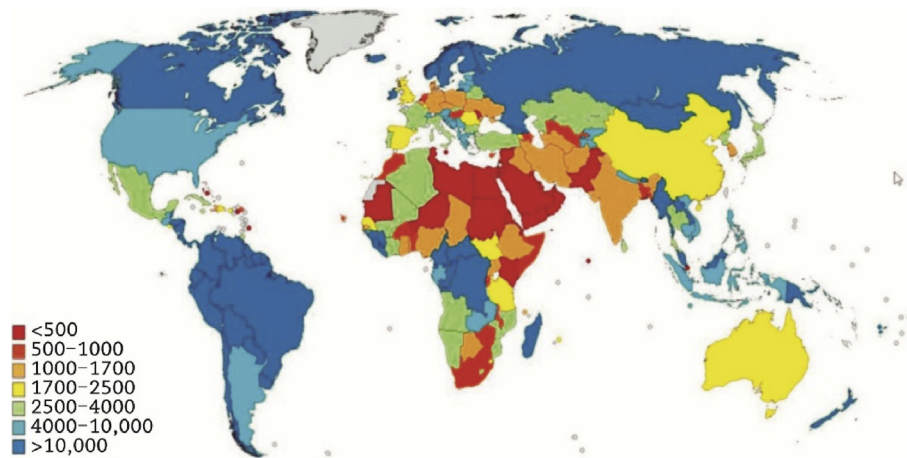


Fig. 2 – Renewable water birthplaces in the world (Hajjari et al., 2017).

burning and eco-friendly renewable energy fuels for CI engine as a replacement for petroleum fuel nowadays is a worldwide consideration (Jasrotia et al., 2019; Krishnamoorthi and Malayalamurthi, 2017). Biodiesel has been found as the most promising alternative fuel in CI engines by many researchers (Agarwal et al., 2015, 2016).

In this review paper, authors discussed the potential of various generations of biodiesel feedstocks from production perspectives. This paper also examined many aspects of these feedstocks which include different biodiesel feedstock sources, biodiesel conversion technology, and second-generation biodiesel performance and emission characteristics. The authors summarized the performance, combustion and emission characteristics of CI engine of hundreds different biodiesels from first, second and third generation of biofuels and compare them with conventional diesel fuel. The review also studies the fatty acid profiles of various biodiesels along with their engine characteristics upon using the surrogate fuels. The review also comprises of production, properties, potential estimation and economics of using biodiesel in compression ignition engines.

2. Biodiesel feedstocks

The characteristics of the diesel engine depend upon the different properties of fuel such as a viscosity of fuel, cloud point, flash point, heating value, fuel density, cetane number, pour point etc. (Bamgboye and Hansen, 2008; Dwivedi et al., 2016; Verma et al., 2019, 2020). The fuel properties should also meet the standard limits assigned by the ASTM, ISO, EN and many more, out of these ASTM is mostly used to compare the fuel properties before its being utilized as a fuel for the CI engine (Arbab et al., 2013). The overall engine performance is superior to the properties of fuel values within the acceptable range of standard limits. There is an improvement in the fuel properties when it is blended with two or more fuels and many researchers have used this combination as a substitute fuel for CI engine (Alptekin and Canakci, 2008; Kim and Lee, 2010). In this section, the properties that are widely used to check the fuel properties

in the CI engine are discussed. The study also evaluates the properties of biodiesel blend with diesel fuel at different blend ratios to improve the fuel quality for CI engine. Therefore, many researchers around the world are studying the quality of biodiesel and its blends concerning blend properties, blending of fuel and storage time (Alptekin and Canakci, 2009; Benjumea et al., 2011). Table 1 shows the different generation's biofuels and their feedstocks.

2.1. Different generations of feedstocks

Biofuels are categorized into three different generations depends upon their feedstock. In this section, a brief discussion on different generations of biodiesel and their feedstocks based on their suitability to use an alternative of fossil fuel has been introduced.

2.1.1. First-generation biodiesel

The first-generation biodiesel is derived directly from food crops, sugarcane, vegetable oil, etc. Crops such as sugar and wheat are widely used as a feedstock for the production of ethanol, while edible oilseeds have been used by many researchers for the production of biodiesel. Some biodiesel from the first generation has its benefits for reducing the greenhouse effect and CO₂ emission, while on the other side, some biodiesel produces an adverse effect. The major problem faced by the first-generation biodiesel is "fuel vs. food" security. Mostly, biodiesel is produced from food crops, and the rise in demand for biofuels has led to increasing the volume of crops that are being diverted from the food market, which results in an increase in food price globally over the last couple of years.

2.1.2. Second-generation biodiesel

Second-generation biofuels have been produced to overcome some limitations with first-generation biofuels. The second-generation biofuel is mostly derived from the non-edible seeds, organic waste, wood, waste food crops, etc., to eliminate some problems with the use of first-generation biodiesel. These are seen as a potential replacement for conventional edible food crops because they do not compete with food crops and do not need land clearing. Several researches

Table 1 – Generations of biofuels and their feedstocks (Altikriti et al., 2015; Fadhil et al., 2016b; Rajak and Verma, 2018a; Rajak et al., 2018).

Generation	Feedstock	Example
First-generation feedstock	Vegetable oil, corn sugar, edible oil seeds, animal fats, food crops, etc.	Palm, soybean, rapeseed, peanut, sunflower oil, coconut seed oil, linseed, castor, groundnut, corn oil, safflower oil, rice bran oil, camphor oil, etc.
Second-generation feedstock	Non-edible oil seeds, forest residues, sugarcane bagasse, etc.	Jatropha curcas, Karanja, mahua, cotton seed neem, annona methyl ester, aphanamixis polystachyn methyl ester, canola, calophyllum inophyllum methyl ester, sea mango, etc.
Third-generation feedstock	Algae, waste oil, waste animal fats	Microalgae oil, bael oil, waste cooking oil, waste frying oil, grease oil, fish oil, etc.

(Altikriti et al., 2015; Can et al., 2017; Fadhil et al., 2018; Ogunkunle and Ahmed, 2019) examined and discussed many aspects of second-generation biodiesel feedstocks. These factors include different biodiesel feedstock sources, biodiesel conversion technology, and second-generation biodiesel performance and emission characteristics.

2.1.3. Third-generation biodiesel

The third-generation biofuel is based on the improvement in the production of biomass. The primary source of the third-generation biofuel is the algae, and it takes many advantages to the specially engineered energy crop. The advantage of using the algae is due to its high energy efficiency, low-cost, and it is a renewable energy feedstock. It is also predicted that the potency of the algae is to generate more and more energy per acre than the conventional food crop. It is unsuitable for food production because algae can grow on land and water, therefore it reduces strain because already there is a depletion of water sources.

2.2. Biodiesel production from oils

Most researchers have focused on the transesterification technique for the production of biodiesel from various sources such as vegetable oil, fats, edible and non-edible seed to reduce the molecular weight of raw oil and viscosity (Puhan et al., 2009; Zareh et al., 2017). In the transesterification process, oil reacts with the alcohol in the presence of strong catalysts resulting in alkyl ester (biodiesel) and a by-product known as glycerol (Fadhil and Ahmed, 2016; Fadhil et al., 2018). The presence of a catalyst is necessary for the production of biodiesel so as to fasten reaction rate and also increase in ester yield. The most commonly used alcohols are ethanol and methanol, whereas as a catalyst, potassium hydroxide (KOH) and sodium hydroxide (NaOH) was used (Fadhil and Saeed, 2016). Table 2 shows the properties of various feedstock used for biodiesel.

3. Diesel engine characteristics

3.1. Combustion characteristics

In order to evaluate the combustion characteristics in a compression ignition engine fueled with biodiesel blends, several researchers chosen the various important parameters such as cylinder pressure, ignition delay, heat release rate the premixed combustion duration, and the fraction of heat release in premixed combustion phase (Chen et al., 2019, 2020; Lodi et al., 2020; Zhang et al., 2020). Start of combustion represented by the point where the heat release rate is zero. The difference between the injection timing and the start of combustion is called the ignition delay. The effect of these parameters are discussed in the preceding section.

3.1.1. Cylinder pressure

Cylinder pressure is the amount of pressure generated inside the combustion chamber on burning the fuel. The combustion process for diesel fuel begins later because of the longer ignition delay compared to biodiesel fuel and its blends. Qi et al. (2017)

Table 2 – Properties of various feedstock biodiesel.

Sr. No.	Biodiesel	CV (MJ/kg)	V (cSt at 40 °C)	MF	MW (kg/kmol)	Density (kg/m ³)	CN	FP (°C)	FPT (°C)	CP (°C)	PP (°C)	Percent of C (%)	Percent of H (%)	Percent of O (%)	Ref.
1	Annona methyl ester	41.100	4.2000	—	—	880.2000	52.00	76.0	92.0	—	—	—	—	10.80	Ramalingam et al. (2016); Senthil and Silambarasan (2015)
2	Aphanamixis polystachyn methyl ester	39.960	4.7177	—	—	873.5000 at 40 °C and 893.0000 at 15 °C	44.00	188.5	—	8.00	8.00	—	—	—	Senthil and Silambarasan (2015)
3	Aegle marmelos oil	36.300	24.3000	C ₁₈ H ₃₆ O ₂	—	896.0000	51.70	—	298.0	—	–5.00	—	—	—	Krishnamoorthi and Malayalamurthi (2017)
4	Butanol	33.630	2.2200	C ₄ H ₉ OH	74.12	794.0000–810.0000	25.00	35.0–37.0	—	—	—	64.86	13.50	21.00	Nabi et al. (2017); Şahin and Aksu (2015); Zheng et al. (2015)
5	Canola biodiesel	39.000	4.4000	C ₁₉ H ₃₅ O ₂	295.50	884.4000	54.30	177.6	—	–8.00	—	—	—	—	Ozsezen et al. (2009)
6	Calophyllum inophyllum methyl ester	38.900	5.7600	—	—	872.0000	58.70	179.0	198.0	13.20	4.00	0.03	—	—	Kshirsagar and Anand (2017); Ramesh Babu et al. (2017)
7	Ethanol	26.800	1.4000	C ₂ H ₅ OH	46.07	785.0000 at 20 °C	5.00	13.0	—	<–26.00	—	52.20	13.00	34.80	Herreros et al. (2015); Labeckas et al. (2014); Morsy (2015)
8	Waste fish oil	41.000	4.1420	—	—	870.0000 –880.0000 at 15 °C	51.50	164.0–173.0	125.0	—	—	82.06	8.64	9.30	Fadhil and Ahmed (2016); Gharehghani et al. (2017); Gnanasekaran et al. (2016)
9	Cotton seed oil	39.500	35.7000	C ₁₈ H ₃₂ O ₂	—	929.0000 at 15 °C	45.00	230.0	228.0	—	—	77.39	11.90	11.10	Daho et al. (2013); Srihari et al. (2017); Singh et al. (2018a, b)
10	Soybean methyl ester	39.700	4.7000	C _{18.74} H _{34.43} O ₂	—	880.0000 at 20 °C	51.00	69.0	—	–2.00	–7.00	—	—	—	Gonca and Dobrucali (2016); Rajak and Verma (2018a, b); Senthil Kumar et al. (2015)
11	Coconut methyl ester	37.800	27.6400	CH ₃ (CH ₂) ₈ –COOH	217.00	0.9089	60.00	391.0	—	279.00	264.00	—	—	—	Singh et al. (2018a, b); Zareh et al. (2017)
12	Hexanol biofuel	39.100	3.4120	C ₆ H ₁₃ OH	102.18	821.8000	23.00	59.0	—	—	—	—	—	15.70	De Pours et al. (2017)
13	Honge oil	36.100	5.6000	—	—	890.0000	42.00	163.0	—	–2.00	1.00	—	—	—	Khandal et al. (2017)
14	Jatropha oil methyl esters	39.500	4.1200	C ₁₂ –C ₂₂	—	—	48.13	165.0	—	–4.00	–6.00	—	—	12.25	Dubey and Gupta (2017); Singh et al. (2018a, b)
15	Waste cooking oil	38.200	5.8700	C _{6.41} H _{12.21} O _{0.68}	—	844.0000	53.00	130.0	132.0	—	—	80.21	12.60	7.30	Asokan et al. (2019); Lapuerta et al. (2005)

16	Corn oil	38.400	4.5000	$C_{17.89}H_{32.8}O_2$	—	880.0000	52.50	165.0	—	−3.00	−5.10	76.70	11.52	10.9	Singh et al. (2018a, b); Shehata et al. (2015)
17	Jojoba oil	42.760	25.4800	—	—	833.0000	52.50	292.0	—	—	6.00	81.80	5.20	—	Huzayyin et al. (2004); Saleh and Selim (2017)
18	Karanja methyl ester	37.300	27.8400	—	—	913.0000	39.00	205.0	209.0	13.20	6.40	—	—	—	Bajpai et al. (2009)
19	Pine oil	42.800	1.3000	$C_{10}H_{16}+$ $C_{10}H_{18}O$	140.00	875.1000	11.00	52.0	—	−12.00	−12.00	—	—	—	Vallinayagam et al. (2014a, b)
20	Kerosene biodiesel	41.780	1.1500	$C_{12}H_{26}$	170.33	797.0000	41.40	40.0	—	−78.00	—	84.61	15.39	—	Patil and Thipse (2015); Roy et al. (2014)
21	Rapeseed methyl ester	37.600	4.5000	$C_{19}H_{35.2}O_2$	—	920.9000	37.60	280.0	—	−4.00	−31.70	—	—	10.90	Senthil Kumar et al. (2015)
22	Palm oil	38.840	4.4000	$C_{18.07}H_{34.93}O_2$	284.17	870.0000	56.50	70.6	—	10.00	—	—	—	—	Bora and Saha (2015); Ozsezen et al. (2009)
23	Pongamia oil methyl ester	39.140	5.5000	$C_{17.69}H_{32.94}O_2$	—	890.0000	55.40	175.0	—	—	4.00	—	—	—	Bora and Saha (2015)
24	Neem methyl ester	41.000	4.6300	$C_{35}H_{44}O_{16}$	720.00	871.0000	53.50	222.0	210.0	−4.00	8.00	—	—	11.00	Dhamodaran et al. (2017); Sharma et al. (2009)
25	Rice bran oil	39.540	3.2400– 4.9800	$C_{18.05}H_{34.9}O_2$	—	880.0000	56.30	152.0	159.0	3.00	−1.00	—	—	11.25	Bora and Saha (2015); Chhabra et al. (2017); Dhamodaran et al. (2017)
26	Linseed methyl ester	39.750	4.2000	—	—	921.0000	48.00	161.0	—	—	−18.00	78.14	9.98	11.72	Puhan et al. (2009)
27	Cynara methyl ester	—	4.8000	$C_{18.7}H_{34.68}O_2$	292.64	887.0000	59.00	—	—	—	—	—	—	10.93	Lapuerta et al. (2005)
28	Mahua methyl ester	36.000	24.5800	—	—	920.0000	57.00	232.0	—	15.00	10.00	77.90	12.80	9.30	Acharya et al. (2017)
29	Kapok methyl ester	39.400	4.2000	—	—	875.0000 at 40 °C	51.00	148.0	—	4.00	−4.40	—	—	—	Senthil Kumar et al. (2015)
30	Turkey rendering fat biodiesel	40.680	4.4900	—	—	885.8000 at 15 °C	52.40	178.1	—	—	4.00	—	—	—	Emiroğlu et al. (2018)
31	Sand apple ethyl ester	—	3.9000	—	—	899.0000	54.30	165.5	—	4.78	1.05	—	—	—	Ogunkunle and Ahmed (2019)
32	Sunflower methyl ester	—	4.1300	$C_{18.9}H_{34.8}O_2$	294.58	885.4000	56.40	—	—	—	—	—	—	10.86	Lapuerta et al. (2005)
33	n-pentanol	34.650	2.8900	$C_5H_{11}-OH$	88.15	814.8000 at 15 °C	20.00	49.0	—	—	—	68.13	13.61	18.15	Dhanasekaran et al. (2017)
34	Terminalia oil	39.220	5.1700	—	—	882.8000	53.00	90.0	—	—	6.00	—	—	—	Chakraborty et al. (2009); Kakati and Gogoi (2016)
35	Bitter almond oil	—	26.1200	—	—	0.9224 at 15.6 °C	—	185.0	—	−4.00	−7.00	—	—	—	Fadhil et al. (2016a, 2017)
36	Bitter almond oil	—	26.1200	—	—	0.9224 at 15.6 °C	—	185.0	—	−4.00	−7.00	—	—	—	Fadhil et al. (2016a)

(continued on next page)

61	Marine fish oil	41.370	7.2000 at 40 °C, 4.4000 at 70 °C	—	—	—	50.90	103.0	—	—	—	—	—	7.19	Lin and Li (2009)
62	Honne oil methyl ester	38.550	5.2000	—	—	—	52.03	170.0	182.0	4.00	—	—	—	—	Channapattana et al. (2015)
63	Dairy washed milk-scum	40.115	3.8000	—	—	875.0000	—	129.0	—	17.00	10.00	—	—	—	Srikanth et al. (2017)
64	Microalgae chlorella protothecoides	40.040	4.2000	—	—	900.0000 at 15 °C	52.00	124.0	—	—	—	—	—	—	Al-lwayzy and Yusaf (2017)
65	Pistacia lentiscus	40.000	3.4400	—	—	860.0000	53.94	168.0	—	—	—	76.17	13.93	9.90	Khiari et al. (2016)
66	Nahar oil	42.330	6.2000	—	—	598.0000	54.00	112.0	—	—	3.00	—	—	—	Kakati and Gogoi (2016)
67	Soya soap stock-based acid oil biodiesel	39.550	5.2000	—	—	886.0000	54.30	102.0	—	—	−8.00	—	—	—	Tarabet et al. (2014)
68	Eucalyptus biodiesel	—	2.9900	—	—	896.0000	53.00	105.0	—	—	—	77.00	12.00	10.32	Tarabet et al. (2014)
69	Mango seed biodiesel	40.400	4.7300	—	—	882.0000	54.00	135.0	—	—	—	—	—	—	Velmurugan and Sathiyagnanam (2016)
70	Animal fat biodiesel	—	4.4200	C ₁₈ H ₃₆ O ₂	—	871.0000	57.49	161.5	—	—	—	81.11	11.67	6.89	Barrios et al. (2014)
71	Tung oil	39.510	115.0000	—	—	945.0000	—	—	—	—	—	—	—	—	Qi et al. (2017)
72	Wood pyrolysis oil	—	25.3000	—	—	—	25.00	98.0	108.0	10.00	2.00	—	—	—	Prakash et al. (2013)
73	Ceiba pentandra biodiesel	40.490	4.1600	—	—	876.9000 at 15 °C	57.20	156.5	—	3.00	2.50	78.00	12.00	11.68	Ong et al. (2014)
74	Safflower oil	38.120	5.8000	—	—	888.0000	—	148.0	—	−5.00	—	59.50	—	—	İlkiliç et al. (2011)
75	Tea seed oil biodiesel	37.512	4.9500	—	—	884.0000 at 20 °C	52.00	>120.0	—	—	—	—	—	—	Serin and Yıldızhan (2018)
76	Chinese Pistache biodiesel	38.265	4.4000	—	942.10	877.0000 at 20 °C	46.00	—	—	—	—	—	—	—	Huang et al. (2010)
77	Salvia macrosiphon oil	—	4.5000	—	856.40	940.0000	48.00	185.0	—	—	—	—	—	—	Hoseini et al. (2017)
78	Sesame oil methyl esters	38.800	5.3400	—	—	882.0000	—	170.0	—	—	—	—	—	—	Banapurmath et al. (2008)
79	Tomato seed oil	41.000	28.0000	—	—	915.0000	—	189.0	—	−8.90	−16.10	—	—	—	Giannelos et al. (2005)
80	Peanut methyl ester	38.700	4.7000	—	—	880.0000	53.21	—	—	—	5.00	—	—	—	Tosun et al. (2014)
81	Carbon black	32.860	2.7400	—	—	860.0000	49.00	56.0	—	—	−12.00	84.40	2.86	1.00	Wamankar et al. (2015)
82	Croton methyl ester	39.786	4.0700–4.1200	—	—	870.4000 at 40 °C	42.40	190.0	—	—	−5.00	76.88	11.04	—	Ruhul et al. (2017)
83	Eucalyptus	45.760	1.5700–2.000	—	—	895.5000 at 40 °C	49.00	67.0	—	—	—	86.00	14.00	—	Srinivas et al. (2017); Tarabet et al. (2014)
84	Paradise oil methyl ester	40.285	5.4000	—	—	875.2000 at 40 °C	53.60	141.2	150.0	—	2.00	—	—	—	Devan and Mahalakshmi (2009)
85	Moringa methyl ester	40.050	5.0500–0.0730	—	—	859.7000 at 40 °C	56.00	176.0	180.5	19.00	19.00	—	—	—	Rashed et al. (2016)

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Table 2 – (continued)

Sr. No.	Biodiesel	CV (MJ/kg)	V (cSt at 40 °C)	MF	MW (kg/kmol)	Density (kg/m ³)	CN	FP (°C)	FPT (°C)	CP (°C)	PP (°C)	Percent of C (%)	Percent of H (%)	Percent of O (%)	Ref.
86	Polanga	41.390	3.9900– 5.3400	–	–	869.0000	57.30	140.0		13.20	4.20	–	–	–	Sahoo and Das (2009)
87	Tobacco seed oil methyl ester	38.430– 39.810	3.5000– 4.2300	–	–	860.0000– 888.5000 at 40 °C	49.00– 51.60	152.0– 165.4	–	–	–	–	–	–	Usta et al. (2011)
88	Waste tire pyrolysis	42.080	2.7100	–	–	887.2000 at 15 °C	44.00	39.0	–	–	–	85.49	10.48	2.74	Ayoob and Fadhil (2020); Wang et al. (2016)
89	Black soldier fly larvae	–	5.2000	–	–	872.0000	50.00	121.0	–	–	–	–	–	–	Nguyen et al. (2017)
90	Ceibapentandra oil	43.600	3.2200	–	–	883.5000	47.00	36.0	–	–	–	–	–	–	Tamilselvan et al. (2018)
91	Ethyl ester of fish oil	40.050	4.7400	–	–	885.0000	52.60	114.0	125.0	–	–	–	–	–	Sakthivel et al. (2014)
92	Crude rice bran oil methyl ester	38.850	4.0300	–	–	–	–	169.0	–	–	–	–	–	–	Saravanan et al. (2013)
93	Spent coffee ground	–	4.6100	–	–	878.0000	–	–	–	12.50	–	–	–	–	Tuntiwiwattanapun and Tongcumpou (2018)
94	Chicken feather biodiesel	–	3.9000	–	–	810.0000	50.00	120.0	–	2.00	–	–	–	–	Purandaradas et al. (2018)
95	Rooster feather biodiesel	–	4.3000	–	–	870.0000	55.00	126.0	–	3.00	–	–	–	–	Purandaradas et al. (2018)
96	Shea olein	46.200	4.4200	–	–	877.0000	–	131.0	–	–	–	–	–	–	Mutyalu et al. (2018)
97	Kaner seed oil	42.400	4.2000	–	–	870.0000	–	175.0	186.0	–	–	–	–	–	Yadav et al. (2017)
98	Cleome viscosa methyl ester	39.200	5.4100	–	–	896.0000	–	178.0	–	–	–	–	–	–	Perumal and Ilangkumaran (2018)
99	Poultry litter oil methyl ester	29.000	5.4800	–	–	737.0000	61.00	154.0	178.0	–	–	–	–	–	Ramesh et al. (2018)
100	Simarouba oil methyl ester	38.500	4.6800	–	–	865.0000	56.00	165.0	172.0	–	14.20	0.10	–	–	Mishra et al. (2018)

experimentally investigated soybean biodiesel and its blends (30%, 50%, and 80%) at variable load conditions in a single-cylinder 4-stroke CI engine. Combustion of biodiesel and its blends produced higher peak pressure at lower engine loads, while at higher load conditions there was no significant change. The cylinder pressure for biodiesel fuel and diesel was found to be 8.12 and 8.11 MPa at 90% load condition, respectively.

3.1.2. Heat release rate

Heat release rate plays a crucial role in the analysis of combustion in CI engine. The model used to calculate the HRR was based on the theories and different laws of thermodynamics. The higher viscosity and low volatility of the rapeseed seed oil biodiesel and its various blends will affect improper fuel atomization and reduces the heat release rate (Atabani et al., 2012). This is because of the high viscosity and lower calorific value of the JME, and on the other side, turpentine biodiesel has CV higher and the low viscosity, and this makes the lake of JBD (Dubey and Gupta, 2018).

3.1.3. Ignition delay

The ignition delay in the CI engine is described by the time interval between the fuel injection timing and the start of combustion. Ignition delayed due to the physical processes (such as atomization, evaporation and mixing of fuel with air) and chemical processes such as pre-combustion reactions. These processes are mainly influenced by fuel characteristics, operating variables and engine design. Ignition delay plays a crucial role in the combustion and emission characteristics of the CI engine fueled with diesel and biodiesel blends.

Dhamodaran et al. (2017) investigated three different biodiesels, namely rice bran, neem and cotton seeds oil as an alternative fuel in CI engine under various loading conditions. They found that RBME indicates low ignition delay because of higher cetane value RBME biodiesel (Kaisan et al., 2017). Ignition delay was higher with diesel fuel, while on the other side, lower ignition duration is found with RBME. Ignition delay for other tested fuel reduced with increasing the load, due to the confinement of heat generating in the previous cycle, dilution of exhaust gas temperature and temperature at the combustion chamber wall. Another possible reason for this reduction in ignition temperature at higher load conditions is higher in-cylinder temperature, which increases the fuel vaporization rate, leading to a decrease in ignition delay.

Several researchers proposed empirical relations to estimate the ignition delay period based on fuel compositions and injection conditions. Wolfer presents the most used empirical link for ignition delay (Westlund, 2009).

$$\tau = AP^n \exp\left(\frac{E_a}{R_u T}\right) \quad (1)$$

where A and n are constants, E_a is the fuel's activation energy, P and T are the mean pressure and temperature of the combustor's compressed air, R_u is the universal gas constant.

Another empirical relation to evaluating the ignition delay is proposed by the Saravanan et al. (2014), which is also applicable to the diesel and biodiesel fuel blends, given below in Eq. (2).

$$\tau = \exp(-(0.0187CN + 8.551))P_i^{0.7352} \exp\left(\frac{4626.44}{T_L}\right) \quad (2)$$

where P_i is the pressure at the start of the injection, T_L is the load condition temperature and CN is the cetane number.

3.1.4. Combustion duration

Behçet et al. (2014) investigated fish oil biodiesel and its blends with diesel fuel in different proportions in a CI engine under variable loading conditions with constant engine speed. The author observed that the combustion duration process increases with an increasing load for all blends of biodiesel because, at higher loading condition, more fuel is injected into the combustion chamber. Blends of biodiesel and neat biodiesel indicated lower combustion duration for all engine operating conditions. The presence of higher oxygen percentage in the biodiesel reduces combustion duration and improves burning at a faster rate (Candeia et al., 2009; Lin and Lin, 2007; Wu et al., 2009).

3.2. Performance characteristics

3.2.1. Brake thermal efficiency

Savariraj et al. (2011) investigated mahua biodiesel and its blends (B25, B50, B75, B100) under different load conditions. They found that the brake thermal efficiency (BTE) for all load conditions is lower for biodiesel and blends. However, increasing biodiesel percentage in diesel reduces BTE compared to diesel due to the higher density, higher viscosity and low calorific value of the mahua oil biodiesel and its blends (Vijay Kumar et al., 2018).

3.2.2. Brake specific fuel consumption

Baiju et al. (2009) investigated the diesel engine characteristics by using Karanja oil methyl and ethyl esters (KOME and KOEE) as a fuel with different blends and regular diesel fuel and at other load conditions. With an increase in engine load, a reduction in BSFC was reported for all tested fuels. B20 KOME blend indicated lower fuel consumption at full load conditions. Therefore, this decrement in the diesel-biodiesel blend is may be due to the engine takes more fuel to obtain the same output power because of its low calorific value than diesel fuel.

3.2.3. Exhaust gas temperature

Sakthivel et al. (2014) used fish oil biodiesel with its blends in different proportion such as (20%, 40%, 60%, 80%, and 100%) in a single cylinder, with a constant engine speed of 1500 rpm under different loading conditions. EGT for blend B20 (270.4 °C) was found close to diesel fuel (269.2 °C). The mean EGT was about 2.4%, 2.1%, 1.1% and 3.3% for blend B40, B60, B80 and B100 respectively, higher than diesel fuel. The increase in EGT for biodiesel fuel is because higher oxygen percentage improves the combustion process. The low volatility and higher viscosity of biodiesel fuel lead to slow-burning of the fuel and causes an increase in temperature (Behçet et al., 2014). Generally, the fuel having a lower cetane number increases the ignition delay and more accumulate the fuel-air mixing, which also effects on the faster heat release rate at the start of the combustion process, and results in higher temperature (Singh et al., 2018a, b).

3.2.4. Engine torque

How et al. (2018b) investigated calophyllum inophyllum biodiesel as a fuel in medium-duty, four cylinders, four strokes, diesel turbocharged, CI engine at different engine speeds (1500, 2000, 2500, 3000, 3500, 4000 rpm) and torque conditions. The results showed maximum torque at 2000 rpm for all the tested fuels. Higher biodiesel blends reported a reduction in engine torque over the entire testing speed range for all fuels. The reduction was accepted, and also many researchers aligned the same (Habibullah et al., 2014; Liaquat et al., 2013). The reason for this decrement is because of biodiesel fuel having higher viscosity and lower calorific than diesel. Besides, this blend with a higher percentage of biodiesel fuel reduces the engine torque.

3.2.5. Brake power

Brake power is the amount of power generated at the output shaft by the engine. Zareh et al. (2017) studied the performance and emission characteristics using three different biodiesel namely castor oil biodiesel (CAB), coconut oil biodiesel (COB) and waste cooking oil biodiesel (WCB). The results showed that the biodiesel properties such as viscosity, lower and higher heating value influenced the brake power of the CI engine. This reduction of power is due to the biodiesel and its blends having low heating value than diesel (Vieira da Silva et al., 2017).

3.3. Emission characteristics

3.3.1. Smoke emission

Gnanasekaran et al. (2016) investigated the effects of the ethyl ester of fish oil (EEFO) and its blends with different injection timing (21°, 24° and 27° before top dead center) on a direct injection characteristic. Results reported higher average smoke emission by 7% for diesel and 3%–4% for biodiesel blend with 21° compared to 24° before the top dead center. Higher smoke emission for biodiesel compared to diesel fuel was measured due to reduced volatility, higher viscosity and improper fuel and air mixing.

3.3.2. Particulate matter (PM) emission

Nabi et al. (2017) investigated the PM emission on multi-cylinder, high-pressure common rail, and turbocharged engine fueled with regular diesel and diesel-butanol blends. The B10, B20 and B30 blends by volume basis at 10%, 20% and 30% n-butanol respectively with diesel fuel at CR17.5 with different engine load. The PM emissions decreased with n-butanol and maximum reduction of 70% with biodiesel at B6 blend due to associated with the absence of aromatics, sulfur and fuel oxygen in the blends. Herreros et al. (2015) obtained the engine characteristics fuelled by ethanol-diesel blends. The PM emission associated with the soot is a solid carbonaceous and soluble organic material that is adsorbed/condensed hydrocarbons onto the soot particles' surface. The PM emission decreases with higher loads and also affected by exhaust gas recirculation.

3.3.3. Carbon dioxide (CO₂) emission

Ramalingam et al. (2016) investigated annona biodiesel fuel additive using 1, 4-dioxane. The CO₂ emission reduced with additive A20, A20DO5 (A20 with 5 mL 1, 4-dioxane) and A20DO10 (A20 with 10 mL 1, 4-dioxane) as compared to diesel fuel due to higher percentage oxygen and CO₂ emission higher with higher engine load. Palash et al. (2015) investigated aphanamixis polystachya methyl ester with diesel on a DI engine. The blend APME10 found to be higher by 6.21% than diesel due to higher oxygen present within the APME10 and obtained complete combustion than diesel.

3.3.4. Carbon monoxide (CO) emission

Vallinayagam et al. (2014a, b) evaluated the effects of neat pine biodiesel and diesel on direct injection engine characteristics. The experiment was conducted at different inlet air temperatures and constant injection timing with constant engine speed. Authors studied CO emission with different air inlet temperature. They reported lower emission for pine biofuel at 40 °C air inlet temperature higher engine load. Meanwhile, the air inlet temperature of 60 °C shows a reduction by 13.2% lower than diesel fuel at

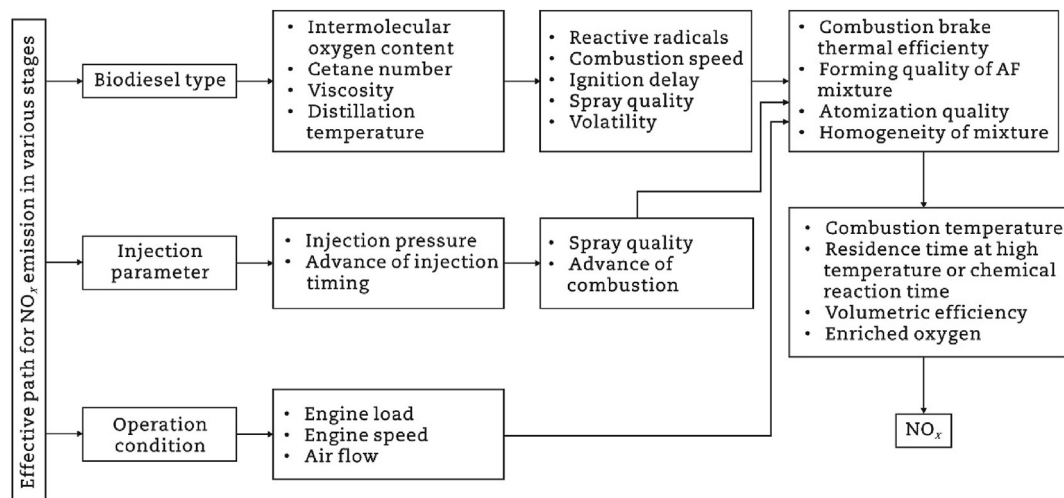


Fig. 3 – Effective parameters and path of NO_x emission during combustion.

Table 3 – Results for different biodiesels on different operating conditions.

Sr. No.	Year	Biodiesel	Fuel blend	Operating condition	Performance	Combustion	Emission	Reference
1	2015	Aphanamixis polystachyn methyl ester	AMPE5, AMPE10	Different engine speed at full load condition	↓: Brake Power ↑: BSFC	–	↓: CO, NO	Palash et al. (2015)
2	2017	Aegle marmelos oil (bael oil)	B20, B30, B40	Varying compression ratio and different engine load	↑: BTE ↓: BSFC	–	↓: CO, HO, NO _x , smoke	Krishnamoorthi and Malayalamurthi (2017)
3	2014	Annona methyl ester	A20, A40, A60, A80, A100	Constant speed and different engine load	↑: BTE	–	↓: CO, HO, smoke ↑: NO _x	Senthil and Silambarasan (2015)
4	2015	n-butanol	nB2, nB4, nB6	Different engine speed and load	↑: BSFC, BTE	↑: cylinder pressure, HRR	↓: CO ₂ , HC, NO _x ↑: smoke	Şahin and Aksu (2015)
5	2015	Butanol	nB100	Constant speed and IMEP	↑: BTE	↑: maximum rate of pressure rises	↓: NO _x , soot	Zheng et al. (2015)
6	2017	Canola biodiesel	5, 10, 15, 20	Constant engine speed with different loading condition	↑: BSFC, BTE	↓: cylinder pressure, HRR	↓: CO, HC, smoke ↑: NO _x , CO ₂	Can et al. (2017)
7	2017	Calophyllum inophyllum methyl ester	CIME10, CIME15, CIME20, CIME25	Different injection timing, injection pressure, and loading conditions	↑: BSFC, BTE	↑: cylinder pressure ↓: HRR	↓: UBHC, CO, SOOT, O ₂ ↑: CO ₂ , NO	Kshirsagar and Anand (2017)
8	2015	Ethanol	25, 50, 75, 100	Different engine speed	↑: BSFC, BTE ↓: EGT	–	↓: NO ↑: CO, HC	Morsy (2015)
9	2017	Waste fish oil	B25, B50, B75, B100	Constant engine speed and different engine load	↑: BTE	↑: cylinder pressure ↓: HRR	↓: CO, UHC ↑: NO _x , CO ₂	Gharehghani et al. (2017)
10	2013	Cotton seed oil	CSO20, CSO40, CSO60, CSO80, CSO100	Different engine load with constant engine speed	↑: BSFC ↓: BTE	↓: HRR	↑: CO ↓: NO _x , CO ₂	Daho et al. (2013)
11	2017	Coconut methyl ester	COB5, COB10, COB20, COB30	Different engine load and engine speed	↑: BSFC, BTE	–	↓: PM, CO, CO ₂ ↑: NO _x	Zareh et al. (2017)
12	2017	Honge oil		Different EGR, fuel injection pressure and injection timing with constant speed and compression ratio	↑: BTE	↑: peak cylinder pressure, HRR ↓: ignition delay, combustion duration	↓: HC, CO, smoke ↑: NO _x	Khandal et al. (2017)
13	2017	Jatropha oil methyl esters	BT50, BT70, and BT90	Constant speed with different engine load	↓: BTE	–	↓: NO _x , CO, HC, smoke ↑: CO ₂	Dubey and Gupta (2017)
14	2004	Jojoba oil	20, 40, 60 and 100	Different engine speed	↑: BTE ↓: BSFC	–	↓: soot ↑: NO _x	Huzayyin et al. (2004)
15	2009	Karanja methyl ester	5,10, 15 and 20	Different engine load	↑: BTE ↓: BSFC, EGT	–	↓: NO _x , smoke	Bajpai et al. (2009)
16	2014	Pine oil	B100	Different engine load	↓: BSFC	↑: cylinder pressure	↓: HC, CO, smoke ↑: NO _x	Vallinayagam et al. (2014b)
17	2014	Kerosene biodiesel	5, 10, 20, 50, 100	Different engine load	↑: BSFC	–	↓: CO, HC, NO _x	Roy et al. (2014)
18	2017	Neem methyl ester	100	Different engine load and constant engine speed	↑: EGT ↓: BTE	↑: HRR ↓: cylinder pressure	↓: CO, UHC ↑: NO _x , smoke	Dhamodaran et al. (2017)
19	2009	Neem biodiesel	20	Different brake power	↑: BTE ↓: BSFC		↓: NO _x , HC, CO, smoke	Sharma et al. (2009)

(continued on next page)

Table 3 – (continued)

Sr. No.	Year	Biodiesel	Fuel blend	Operating condition	Performance	Combustion	Emission	Reference
20	2009	Linseed methyl ester	100	Different injection pressure	↓: BTE	↑: cylinder pressure ↓: ignition delay, combustion duration	↓: CO, HC, CO ₂ ↑: NO _x	Puhan et al. (2009)
21	2015	kapok methyl ester	20, 40, 60, 80, 100	Different engine load and constant engine speed	↑: BSFC, EGT ↓: BTE	–	↓: CO, HC, NO _x	Senthil Kumar et al. (2015)
22	2018	Turkey rendering fat biodiesel	TRFB10, TRFB20, TRFB50	Different engine load and constant engine speed	↑: BSFC ↓: BTE	↑: cylinder peak pressure, HRR	↓: smoke ↑: NO _x	Emiroğlu et al. (2018)
23	2017	Calophyllum inophyllum methyl ester	butylated hydroxytoulene, ethanol	constant engine speed	↑: BSFC, BTE	–	–	Ashok et al. (2017)
24	2013	Camelina sativa oil	100	Different engine speed	↓: BTE	–	↑: NO _x , HC, CO, smoke	Kruczyński (2013)
25	2018	Mustard oil	M10, M20, M30	constant speed with various load condition	↑: BSFC ↓: indicated thermal efficiency	↓: cylinder pressure	↓: CO, smoke, ↑: NO _x	Uyumaz (2018)
26	2016	Koroch seed oil	10, 20	Different engine load	↑: BTE, indicated thermal efficiency	–	↓: smoke	Kakati and Gogoi (2016)
27	2017	Hazelnut methyl ester	20, 50, 100	Different engine speed	↓: BTE	–	–	Efe et al. (2018)
28	2018	Bauhinia variegata oil	10, 20, 30, 40, 100	Different engine load	↓: BTE	–	↓: HC, CO ↑: NO _x	Yatish et al. (2018)
29	2016	Pistacia lentiscus	BD5, BD30, BD50, BD100	Constant engine speed with different engine load	↑: BSFC, BTE	–	↓: HC, CO, PM ↑: NO _x	Khiari et al. (2016)
30	2015	Aphanamixis polystachya methyl ester	APME5, APME10	Variable engine speed	↓: torque, brake power ↑: BSFC	–	↓: CO, HC ↑: CO ₂ , NO	Palash et al. (2015); Wu et al. (2013)
31	2017	Soya soap stock-based acid oil biodiesel	100	Different engine speed	↓: indicated thermal efficiency, BTE	↑: cylinder peak pressure, cylinder temperature, injection duration, combustion duration ↓: heat release rate	↓: UHC, CO, smoke ↑: NO _x	Tripathi and Subramanian (2017)
32	2014	Eucalyptus biodiesel	100	Constant speed with various load condition	↑: BSFC	–	↓: UHC, CO, CO ₂ ↑: NO _x	Tarabet et al. (2014)
33	2014	Animal fat biodiesel	A10, A20, A25, A30, A 40 and A50	Constant speed with various load condition	↑: SFC, BTE	–	↑: NO _x ↓: particle number	Barrios et al. (2014)
34	2017	Salvia macrosiphon oil	5, 10, 15 and 20	Different engine speed	↑: BTE ↓: SFC	–	↓: CO, HC ↑: CO ₂ , NO _x	Hoseini et al. (2017)
35	2017	Croton methyl ester	CM20	Constant speed with various load condition	↑: BSFC, EGT ↓: BTE	↓: ignition delay, combustion duration, peak heat release rate	↓: CO, HC ↑: NO _x	Ruhul et al. (2017)
36	2009	Paradise oil methyl ester	20, 30, 40, 50	Different engine loading condition	↑: BTE	–	↓: CO, HC, CO ↑: NO _x	Devan and Mahalakshmi (2009)
37	2011	Tobacco seed oil methyl ester			NA	–	–	Usta et al. (2011)

38	2015	Black soldier fly larvae				—	—	Nguyen et al. (2017)
39	2014	Ethyl ester of fish oil	20, 40, 60, 80 and 100	Constant speed and variable engine load	↑: BTE	↓: ignition delay, peak pressure, rate of pressure rises	↓: HC, CO, NO _x ↑: smoke	Sakthivel et al. (2014)
40	2018	Chicken feather biodiesel	NA	NA	NA	—	—	Purandaradas et al. (2018)
41	2016	Kaner seed oil	20, 40 60	Different compression ration with variable load	↑: BSFC ↓: BTE	—	—	Yadav et al. (2017)
42	2009	Waste palm oil	100	Constant rpm and full load condition	↓: BTE	↑: cylinder pressure ↓: ignition delay	↓: HC, CO, smoke ↑: NO _x	Dwivedi et al. (2018); Ozsezen et al. (2009)
43	2015	Kerosene biodiesel	5, 10 and 15	Different engine load and constant engine speed	↑: BSFC ↓: BTE	—	↓: NO _x ↑: HC, CO, smoke	Patil and Thipse (2015)
44	2016	Annona biodiesel	A20, A20DO5, A20DO10	Constant speed and different brake power	↑: BSFC, BTE	—	↓: HC, CO, NO _x , smoke	Ramalingam et al. (2016)
45	2015	Ethanol	5E, 10E, 15E	Different load and IMEP	↑: EGT	—	↓: soot, NO _x , PM	Herreros et al. (2015)
46	2014	Ethanol	5E, 10E, 15E	Different engine speed	↑: BSFC ↓: BTE	↑: HRR ↓: cylinder pressure	↓: CO ↑: NO _x , HC	Labeckas et al. (2014)
47	2016	Fish oil	B20, B40, B60, B80, B100	Constant speed and different engine load	↑: BTE	↓: HRR, cylinder pressure	↓: CO, HC, NO _x	Gnanasekaran et al. (2016)
48	2017	Cotton seed oil	DBD-1, DBD-2, DBD-3, 80D-20BD	Different brake power	↑: BSFC, BTE	↑: cylinder pressure	↓: NO _x , HC, CO, smoke	Srihari et al. (2017)
49	2016	Soybean methyl ester	50, 100	Different engine load with constant engine speed	↑: BTE ↓: power	—	—	Gonca and Dobrucali (2016)
50	2017	Hexanol biofuel	HEX10, HEX20, HEX30	Different EGR and injection timing	NA	↑: cylinder pressure, HRR	↓: CO, smoke ↑: NO _x , HC	De Pours et al. (2017)
51	2015	Corn oil	C20	Different engine load, engine speed, and injection pressure	↑: BTE ↓: BSFC	↑: cylinder pressure	—	Shehata et al. (2015)
52	2017	Jojoba methyl ester	B5, B10, B15, B20, B25	Constant speed with different engine load	↑: BTE ↓: BSFC	—	↓: HC, CO, smoke ↑: NO _x	Saleh and Selim (2017)
53	2014	Pine oil	40, 50, 60,70	Different brake power	↓: BTE	↓: cylinder pressure, HRR	↓: CO, smoke	Vallinayagam et al. (2014a)
54	2015	Palm oil	100	Different engine load and constant engine speed	↑: BTE ↓: BSFC	—	↓: CO, HC ↑: NO _x	Bora and Saha (2015)
55	2017	Rice bran oil	100	Different engine load and constant engine speed	↑: BTE ↓: BSFC	↑: cylinder pressure ↓: ignition delay	↑: NO _x , HC	Chhabra et al. (2017)
56	2017	n-pantanol	5, 10, 20	Different EGR with constant engine speed	↑: BSFC ↓: BTE	—	↓: NO _x , smoke	Dhanasekaran et al. (2017)
57	2015	Beauty leaf	B5, B10	Different engine speed	↑: BSFC, EGT ↓: BTE	—	↓: NO _x , HC, CO, smoke	Hassan et al. (2015)
58	2012	Cashew nut shell oil	10, 20, 30	constant speed with various load condition	↑: BSFC, EGT ↓: BTE	↑: ignition delay, combustion duration ↓: HRR, MRPR	↓: NO ↑: HC, CO	Kasiraman et al. (2012)
59	2016	Orange oil	10, 20, 30	constant speed with various load condition	↓: BTE	—	—	Kasiraman et al. (2016)

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Table 3 – (continued)

Sr. No.	Year	Biodiesel	Fuel blend	Operating condition	Performance	Combustion	Emission	Reference
60	2017	Castor oil methyl ester	CME20	Different engine speed	↑: power, torque ↓: BSFC	–	↓: CO ↑: NO _x	Ozcanli et al. (2017)
61	2009	Waste plastic oil	100	Different injection timing and engine load	↑: BTE	↓: cylinder pressure	↓: CO, UHC, NO _x	Mani and Nagarajan (2009)
62	2018	Coconut oil biodiesel	B20, B50	Different SOI and with constant engine speed	NA	–	↓: NO _x , smoke	How et al. (2018a)
63	2008	Waste frying oil methyl ester	100	Different engine speed	↓: BTE	–	↓: CO, NO _x ↑: smoke	Utlu and Koçak (2008)
64	2014	Terebinth oil	B10, B50	Different engine speed	↑: BSFC, ↓: power	–	↓: CO, HC, CO ₂ ↑: NO _x	Ilkiliç et al. (2015)
65	2016	Cymbopogon flexuosus biofuel	C20, C100	constant speed with various load condition	↑: BTE	↑: cylinder peak pressure, HRR	↓: NO _x , HC, CO, smoke	Dhinesh et al. (2017)
66	2014	Algae oil methyl esters (AME)	AME20	Different engine load	↑: BTE	–	↓: HC, smoke	Makarevičienė et al. (2014)
67	2009	Tallow methyl ester	B5, B20, B50, B100	Different engine speed	↑: BSFC ↓: BTE	–	↓: CO, NO _x , SO ₂ , smoke	Öner and Altun (2009)
68	2009	Thész-Boros-Király	100	Different engine speed, load, and torque	NA	↑: ignition delay, HRR	↓: HC ↑: NO _x	Öner and Altun (2009)
69	2018	Pithecellobium dulce seed oil	20, 40, 60, 80	Constant speed with various load condition	↑: BSFC, ↓: BTE, EGT	↓: cylinder pressure, HRR	↑: CO ₂ , smoke ↓: CO, HC, NO _x	Chandra Sekhar et al. (2018)
70	2017	Rubber seed oil	25, 50, 75, 100	Constant speed with various load condition	↑: BTE	↑: peak pressure, maximum rate of pressure rises, HRR, ignition delay ↓: combustion duration	↓: smoke ↑: HC, NO _x , CO	Geo et al. (2017)
71	2015	Honne oil methyl ester	20, 40, 60, 80, 100	Difference compression ratio and engine load with constant engine speed	↑: BSFC ↓: BTE	–	↓: CO, HC, smoke ↑: NO _x	Channapattana et al. (2015)
72	2017	Microalgae chlorella protothecoides	MCP20, MCP50, B100	Different engine speed	↓: brake power torque, exhaust gas temperature ↑: BSFC, BTE	–	↓: CO, CO ₂ , NO _x , ↑: O ₂	Al-lwayzy and Yusaf (2017)
73	2015	Mango seed biodiesel	20, 100	Different engine load and ppm	↓: BTE	–	↓: NO _x , ↑: CO, smoke, HC	Velmurugan and Sathiyagnanam (2016)
74	2017	Tung oil	DT30E20, DT30E30	Different engine speed	↑: BSFC, BTE	↑: peak pressure, maximum rate of pressure rises, HRR	↓: smoke, CO, HC ↑: NO _x	Qi et al. (2017)
75	2013	Wood pyrolysis oil	5, 10, 15	Different engine load	↑: BTE	↑: cylinder pressure ↓: ignition delay	↓: NO, smoke	Prakash et al. (2013)
76	2014	Ceiba pentandra biodiesel	10, 20, 30, 50	Different engine speed	↓: BSFC	–	↓: CO ₂ , CO, smoke ↑: NO _x	Ong et al. (2014)
77	2011	Safflower oil	B5, B20, B50	Different engine speed	↑: BSFC	–	↓: PM, smoke ↑: NO _x , HC	Ilkiliç et al. (2011)
78	2017	Tea seed oil biodiesel	B10, B20	Different engine speed	↑: BSFC	–	↓: CO ↑: CO ₂ , NO _x	Serin and Yıldızhan (2018)

79	2010	Chinese pistache biodiese	100	Constant speed with various load condition	↑: SFC	—	↓: CO, HC, NO _x , smoke	Huang et al. (2010)
80	2014	Peanut methyl ester	100	Different engine speed	—	—	↓: CO ↑: NO _x	Tosun et al. (2014)
81	2014	Carbon black	10	Different injection pressure and load	↑: BTE ↓: EGT	—	↓: HC, CO, NO _x , smoke, NO	Tosun et al. (2014)
82	2016	Moringa methyl ester	20	Different engine speed	↑: BSFC ↓: BP	—	↓: CO, HC, ↑: NO	Rashed et al. (2016)
83	2009	Polanga	20, 50, 100	Different engine loading condition	NA	↑: peak cylinder pressure ↓: ignition delay	—	Sahoo and Das (2009)
84	2018	Spent coffee ground	NA	NA	NA	—	—	Tuntiwiwattanapun and Tongcumpou (2018)
85	2016	Shea olein biodiesel	10, 20, 100	Constant load and different injection pressure	↑: BSFC, BTE ↓: EGT	—	↓: CO, HC ↑: NO _x	Mutyalu et al. (2018)
86	2018	Cleome viscosa methyl ester	20, 40, 60, 80, 100	Different engine load	↑: EGT ↓: BSFC, BTE	—	↓: HC, CO, CO ₂ ↑: NO _x	Perumal and Ilangkumaran (2018)
87	2018	Poultry litter oil methyl	20	Different engine load condition	↑: BTE	↓: cylinder pressure, HRR	↓: UHC, CO, NO _x	Ramesh et al. (2018)
88	2016	Cymbopogon flexuosus biofuel	10, 20, 30, 40, 100	constant speed with various load condition	↑: BSFC ↓: BTE	↓: cylinder pressure, HRR	↓: HC, CO, smoke ↑: CO ₂ , NO _x	Dhinesh et al. (2016)
89	2011	Mahua biodiesel	25, 50, 75, 100	Different engine load and constant engine speed	↑: BSFC ↓: BTE	—	↓: HC ↑: NO, CO, smoke	Savariraj et al. (2011)
90	2017	Waste cooking oil	B20, B30, B40, B100	Different engine load condition	↑: BSFC ↓: BTE	↑: HRR ↓: cylinder pressure	↓: CO ↑: NO _x	Senthur Prabu et al. (2017)

Note: “↑” means increase. “↓” means decrease. “—” means no change. “NA” means not available.

100% load due to higher in-cylinder temperature and the essential occurrence of oxygen within pine oil.

3.3.5. Unburnt hydrocarbon (UHC) emission

Saleh and Selim (2017) performed an experimental investigation on characteristics of a diesel engine using diesel fuel, jojoba methyl ester-diesel-ethanol ternary fuel blends (B5, B10, B15, B20, and B25) with variable engine load and constant engine. The single-cylinder direct-injection engine used in this investigation. The investigated result of UHC emission shows that it increased with a higher engine load. This happens due to the rich fuel-air mixture in the combustion chamber and increasing fuel consumption. This shows a reduction in UHC emission for the JME blend of B10 by 39% at 520 kPa and B5 by 27% lower compared to diesel fuel due to more oxygen content. They conclude that UHC emission highest lower recorded at B10 blend compared to diesel fuel.

3.3.6. Oxides of nitrogen (NO_x) emission

In the exhaust of CI engine, NO_x formation occurred due to the thermal NO_x procedure, which is affected by the burned mixture temperature, the confined air/fuel ratio and the dwell time at higher temperatures in the engine cylinder. Chen et al. (2018c) provides a detail discussion on the NO_x emission of biodiesel. Authors reported that fuel injection parameters such as injection timings and injection pressure perform an important role in combustion temperature and NO_x emissions.

Ruhul et al. (2017) investigated the effects of millettia pinnata and croton megalocarpus biodiesel on a DI CI engine characteristics with variable engine load and engine speed. The result of NO_x emission evaluated the used of chain reaction based on zeldovich mechanism. The NO_x emission obtained higher with engine speed and higher for blends biodiesel (MP20, CM20, MP15CM5, MP10CM10 and MP5CM15) compared to diesel fuel due to a higher percentage of oxygen contents and higher combustion temperature within the premixed and diffusion combustion phase. Numerous studies (Chen et al., 2018a; Dwivedi et al., 2019; Geng et al., 2020) have been established the increasing trend of NO_x emission by using biodiesel. But some studies (Guo et al., 2019; Nair et al., 2017; Pereira et al., 2007) also shown the reduction in NO_x due to the addition of biodiesel. Nair et al. (2017) and Pereira et al. (2007) reported less NO_x emission by using neem biodiesel and soybean biodiesel, respectively, than diesel. At low load, Chen et al. (2018b) reported that the NO_x emission of biodiesel is less than that of diesel for the engine operating under lower or medium speeds due to weak air swirl movement and poor in-cylinder heat condition.

NO_x emission of biodiesel is a significantly important issue, and it depends upon various parameters such as biodiesel type, engine technology and structure, operation conditions, combustion temperature and injection parameters (Dwivedi et al., 2018; Verma et al., 2020; Zare et al., 2018, 2020). Effective parameters and path of NO_x emission is shown in Fig. 3. The engine characteristics for various biodiesel fuel blends are reported by researchers are shown in Table 3.

4. Conclusions

First, second and third-generation biodiesel and their blends have different properties that influence diesel engine characteristics such as BTE, BSFC, EGT, smoke emission, NO_x emission, brake power, etc. Therefore, the selection of different generation feedstock is necessary and generally depends upon the domestic birthplaces of the region. The variation with different generation biodiesel feedstocks and its blends properties will impact the overall performance, combustion and emission characteristics of the CI engine. There are numerous previous studies that have been reviewed in this study which can be concluded based on previously studied.

- First-, second- and third-generation biodiesel and their blends feedstocks will impact energy and percentage of oxygen, cetane number, the heating value of fuel, density, and viscosity.
- The calorific value of the fuel is an important attribute to produce engine power. It was observed that different generation biodiesel blends reduce the engine power because of lower calorific values of fuel (MJ/kg) than diesel fuel. For some biodiesel, the calorific value of the fuel is higher such as 42.76, 45.76, 43.60, 46.20, 42.83, and 42.80 MJ/kg for jojoba, eucalyptus, ceibapentandra oil, shea olein, waste tire pyrolysis oil, and pine oil and lower calorific value are found for 26.80, 29.00, 35.10, 34.65 and 30.80 MJ/kg for ethanol, poultry litter oil methyl ester, corn oil, n-pentanol and bisethoxy methyl furan as compared to diesel fuel (42.50 MJ/kg), respectively.
- The combustion quality should be improved to meet the higher engine thermal efficiency; therefore, the maximum amount of fuel energy produced by the fuel is transferred to the engine. Therefore, oxygen is the crucial property for biodiesel that will improve the combustion process and efficiency of the engine. Besides this, preventing heat losses produces during the combustion process and ignition delay is lower due to the higher cetane value of the fuel, and it also improves the air/fuel rate and indicates a better combustion process.
- PM emission is a composite combination of carbon, sulfur, nitrogen and other compounds produced. In numerous previous studies presented that biodiesel-diesel blends (B20) reduce PM emissions by 22% compared to regular diesel fuel, which is occurred due to a higher percentage of oxygen that helps to improve combustion and spray characteristics.
- Unburnt hydrocarbon (UHC) emission indicated the level of inclusiveness for one combustion progression as it measures the total of unburned hydrocarbon. According to many studies, diesel-biodiesel blends reduce the UHC by about 15% for blend B20. Because of the higher cetane value of the blend, it causes a shorter ignition delay period and improves the efficiency of the engine.
- Percentage of oxygen content within the fuel leads to increase NO_x emission, as the presence of oxygen percentage of the biodiesel will form complete combustion of the fuel and it also increases the combustion temperature inside

the combustion chamber. Diesel-biodiesel has shown lower NO_x emissions up to 10% due to the presence of a higher percentage of oxygen for blend B20 compared to diesel fuel.

- Diesel-biodiesel blends (B20) show a reduction in CO emissions by 18% compared to regular diesel fuel. It is noticed that the oxygen percentage in biodiesel plays an essential role in sustaining the oxidation process during the exhaust stroke. The CO emission gets affected by energy contents and latent heat of vaporization of the fuel.

Thus, based on the review, it can be concluded that first-, second- and third-generation biodiesel-diesel blends slightly affected CI engine performance and showed a reduction in destructive exhaust gas emissions fueled with diesel-biodiesel blends of first-, second- and third-generation than with diesel fuel.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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Nomenclature

ANN	Artificial neural network
APME	Aphanamixis polystachya methyl ester
AC	Air cooling
bTDC	Before top dead center
BTE	Brake thermal efficiency
BHT	Butylated hydroxytoluene
BSFC	Brake specific fuel consumption
BME	Butanol-methyl ester
BP	Brake power
CA	Crank angle
CO ₂	Carbon dioxide
CO	Carbon monoxide
CI	Compression ignition
CN	Cetane number
C	Carbon
CFB	Cymbopogon flexuosus biodiesel
CR	Compression ratio
CAB	Castor oil biodiesel
COB	Coconut oil biodiesel
CIME	Calophyllum inophyllum methyl ester
CM	Croton megalocarpus
CP	Cloud point
CV	Calorific value
EGT	Exhaust gas temperature
EGR	Exhaust gas recirculation
EME	Ethanol methyl ester

EEFO	Ethyl ester of fish oil
FT	Fuel type
FP	Flash point
FPT	Fire point
HC	Hydrocarbon
H	Hydrogen
HRR	Heat release rate
HEX	Hexanol
IC	Internal cooling
IP	Injection pressure
JME	Jatropha methyl ester
KV	Kinetic viscosity
MFB	Mass fraction burned
MME	Methanol-methyl ester
MP	Millettia pinnata
MF	Molecular formula
MW	Molecular weight
NO _x	Nitrogen oxide
NG	Natural gas
NH	Nozzle hole
ND	Nozzle diameter
O ₂	Oxygen
PM	Particulate matter
PME	Peanut methyl ester
PP	Pour point
RC	Radiator cooling
RBME	Rice brain methyl ester
SOI	Start of injection
SO _x	Sulfur oxide
TDC	Top dead center
UHC	Unburned hydrocarbon
WPO	Wood pyrolysis oil
WCB	Waste cooking oil biodiesel
WC	Water cooling

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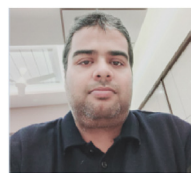
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